Supplementary material for: Retrofitting a commercial RF induction generator into a computer-controlled, vacuum-integrated annealing system for reactive-metal grain growth

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This document collects the supplementary figures (Figs. S1–S10), the specimen–thermal-history linkage (Table S1), the itemized bill of materials (Table S2), and an inventory of the openly available design files (Sec. SVII) that support the main text. The complete raw data and design package is openly available at `https://github.com/vertical-cloud-lab/custom-induction-furnace`, with an archival snapshot deposited at Zenodo, 10.5281/zenodo.20878017. Each run ID in Table S1 links directly to the corresponding raw run log. Alt text accompanies each figure and table caption.

SI Hardware

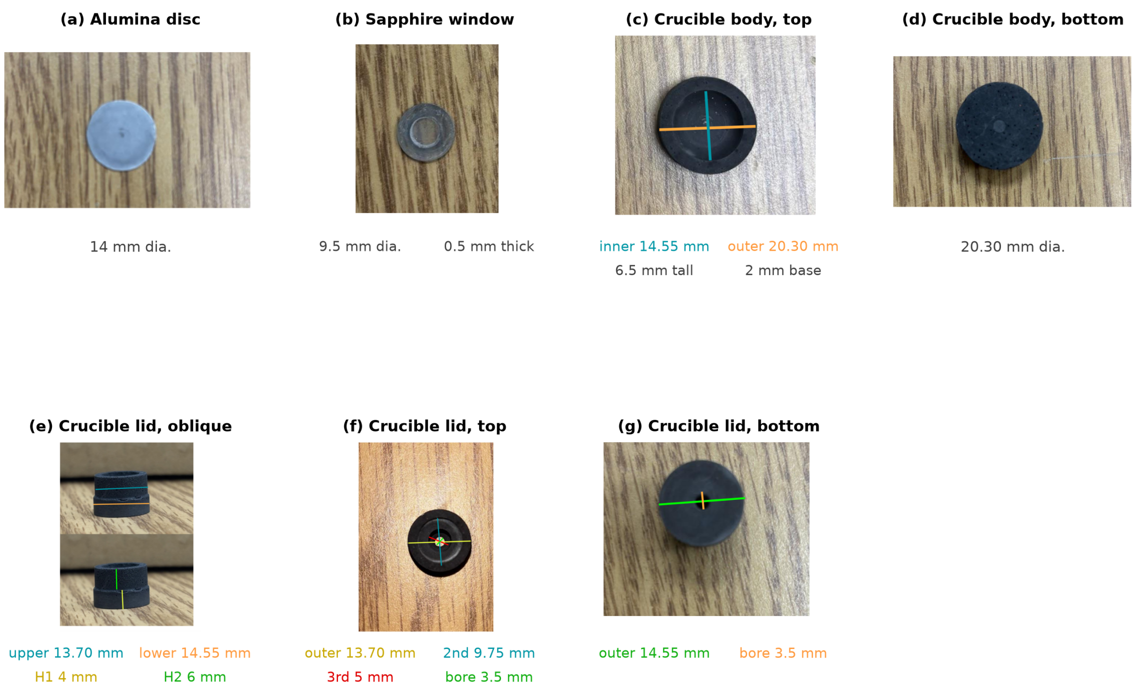

Figure S1: Measured dimensions of each part of the graphite crucible/susceptor assembly, with the coloured measurement lines drawn on the corresponding photograph. (a) Alumina spacer disc, 14 mm dia. (b) Sapphire pyrometer window, 9.5 mm dia., 0.5 mm thick. (c) Crucible body, top: machined sample cavity 14.55 mm ID (teal), 20.30 mm OD (orange); the cup is 6.5 mm tall with a 2 mm base. (d) Crucible body, bottom face, 20.30 mm dia. (e) Crucible lid, oblique: the stepped plug seats into the cavity — 13.70 mm upper step (teal), 14.55 mm lower shoulder (orange); the upper step is 4 mm tall (H1, yellow) and the lower shoulder 6 mm tall (H2, green). (f) Crucible lid, top: four concentric steps of 13.70 mm (outermost), 9.75 mm, 5 mm, and a 3.5 mm central pyrometer-sighting bore. (g) Crucible lid, bottom: 14.55 mm seating face (green) and the 3.5 mm through-bore (orange). Fully assembled the crucible stands 13 mm tall.

Alt text: Seven labeled photographs of the crucible parts, each overlaid with colored measurement lines and numeric callouts giving the diameters, step heights, and the central pyrometer bore dimensions.

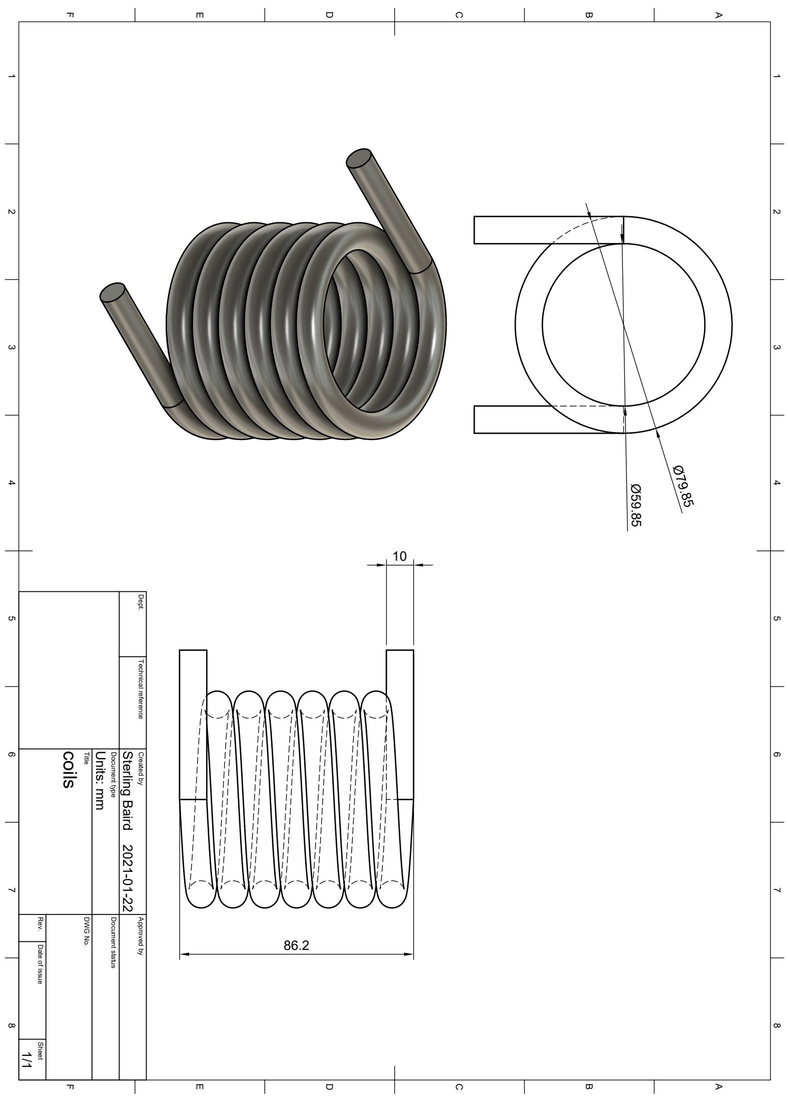

Figure S2: Work-coil fabrication drawing used for this build, showing the coil geometry and dimensional callouts used to form and mount the liquid-cooled copper coil around the quartz chamber.

Alt text: Engineering drawing of the liquid-cooled copper work coil with dimensional callouts for the coil height, inner diameter, number of turns, and lead routing.

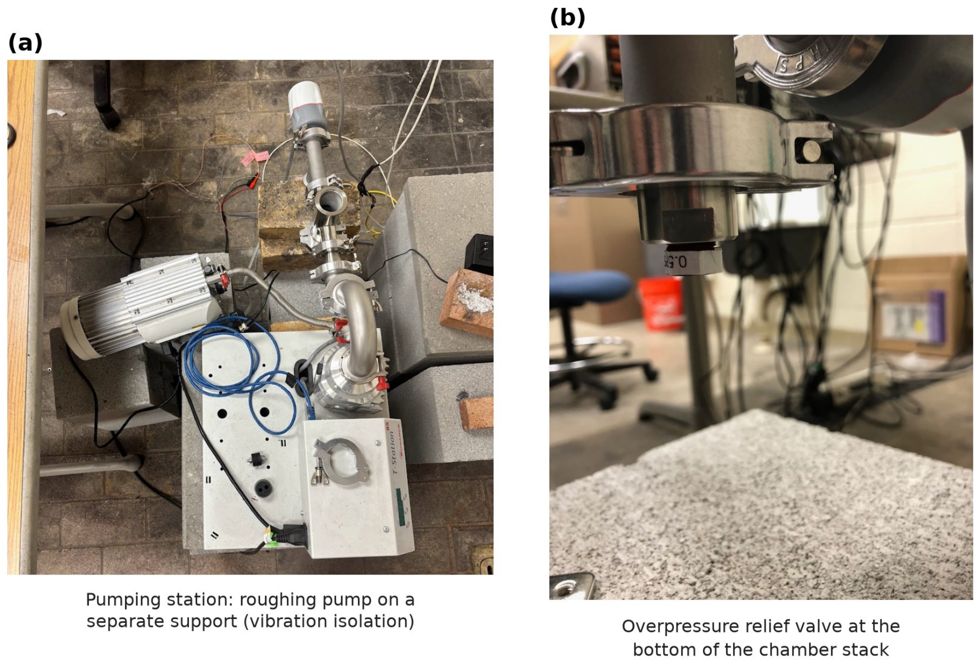

Figure S3: Vacuum and gas-handling details. (a) Turbo pumping station with the roughing pump mounted on a separate support. The pump couples to the pumping station only through a flexible hose, so its vibration is not transmitted into the chamber. (b) Overpressure relief valve (0.5 psi cracking pressure) at the bottom of the vacuum column, below the chamber.

Alt text: Two labeled photographs: the turbo pumping station with its separately supported roughing pump connected by a flexible hose, and the overpressure relief valve mounted at the bottom of the vacuum column.

SII Thermal performance

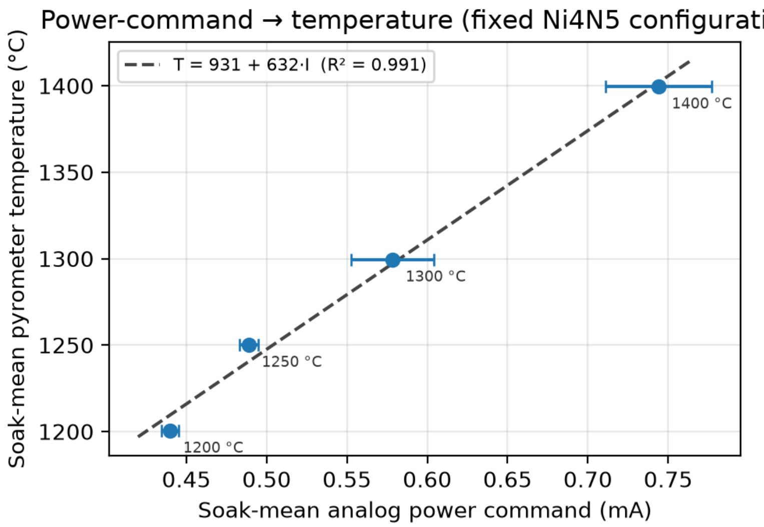

Figure S4: Power-command → temperature calibration for one fixed Ni (4N5, i.e., 99.995 % purity) configuration. Points are soak-mean values; horizontal bars are the in-soak standard deviation of the analog command. The dashed line is the linear operational calibration ($R^2 = 0.991$, $n = 4$).

Alt text: Scatter plot of soak-mean temperature against analog power command with four points between 1200 and 1400 degrees Celsius and a dashed straight fit line through them.

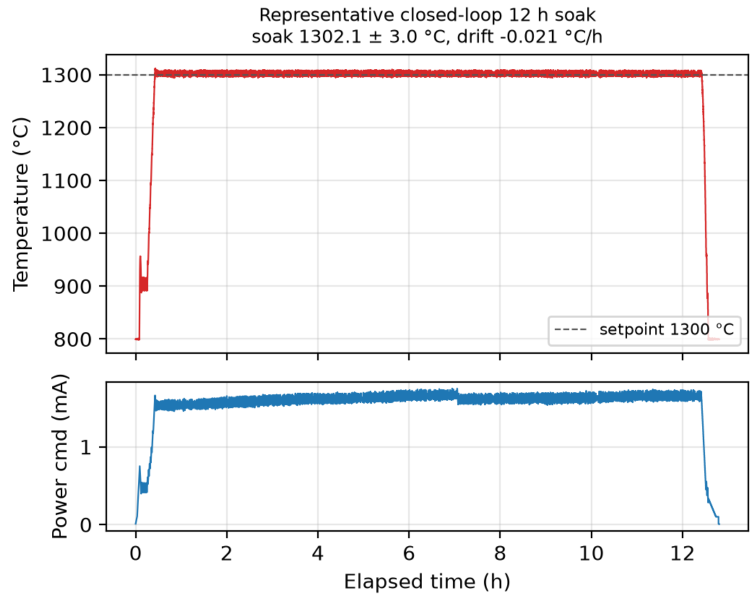

Figure S5: Representative closed-loop anneal (Ni (4N5), nominal 1300 °C / 12 h): pyrometer temperature with setpoint overlay (top) and analog power command (bottom). Soak mean $1302.1 \pm 3.0\,^{\circ}\mathrm{C}$, drift $-0.02\,^{\circ}\mathrm{C/h}$.

Alt text: Two stacked line plots over twelve hours: the pyrometer temperature rising to and holding near 1300 degrees Celsius on its setpoint, and the analog power command settling to a steady value below.

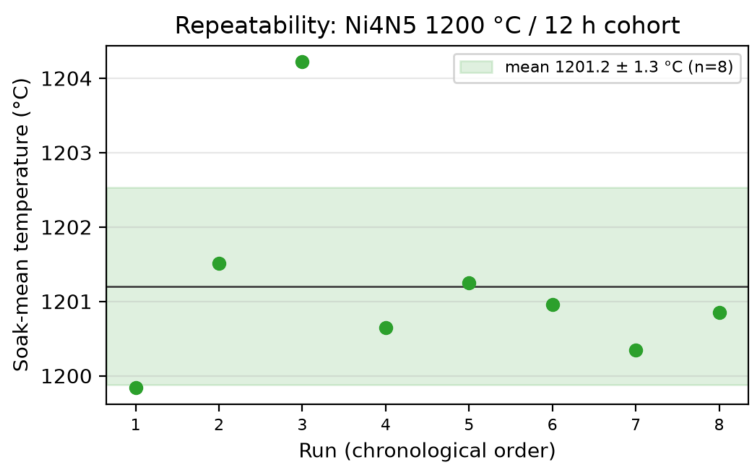

Figure S6: Repeatability of the Ni (4N5) 1200 °C / 12 h cohort ($n = 8$). Points are per-run soak-mean temperatures; the band is mean $\pm$ SD ($1201.2 \pm 1.3\,^{\circ}\mathrm{C}$).

Alt text: Plot of eight per-run soak-mean temperatures clustered near 1200 degrees Celsius inside a shaded band marking the mean plus or minus one standard deviation.

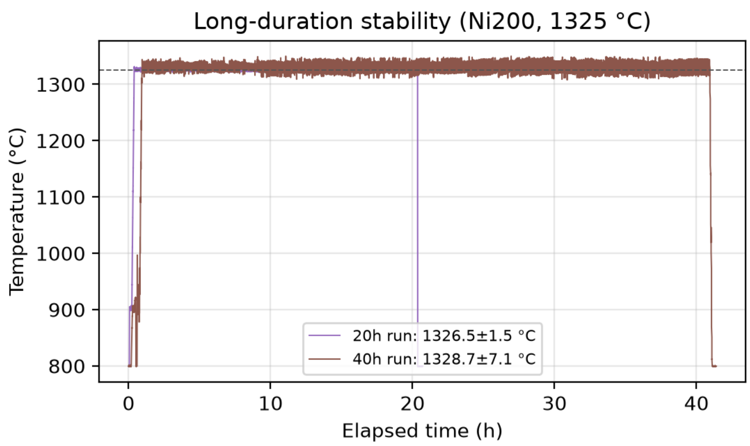

Figure S7: Long-duration thermal stability: Ni200 (commercially pure wrought nickel, UNS N02200) anneals at 1325 °C held for 20 h and 40 h.

Alt text: Line plot of two anneal temperature traces at 1325 degrees Celsius, one held twenty hours and one held forty hours, both essentially flat across the soak.

SIII Microstructure

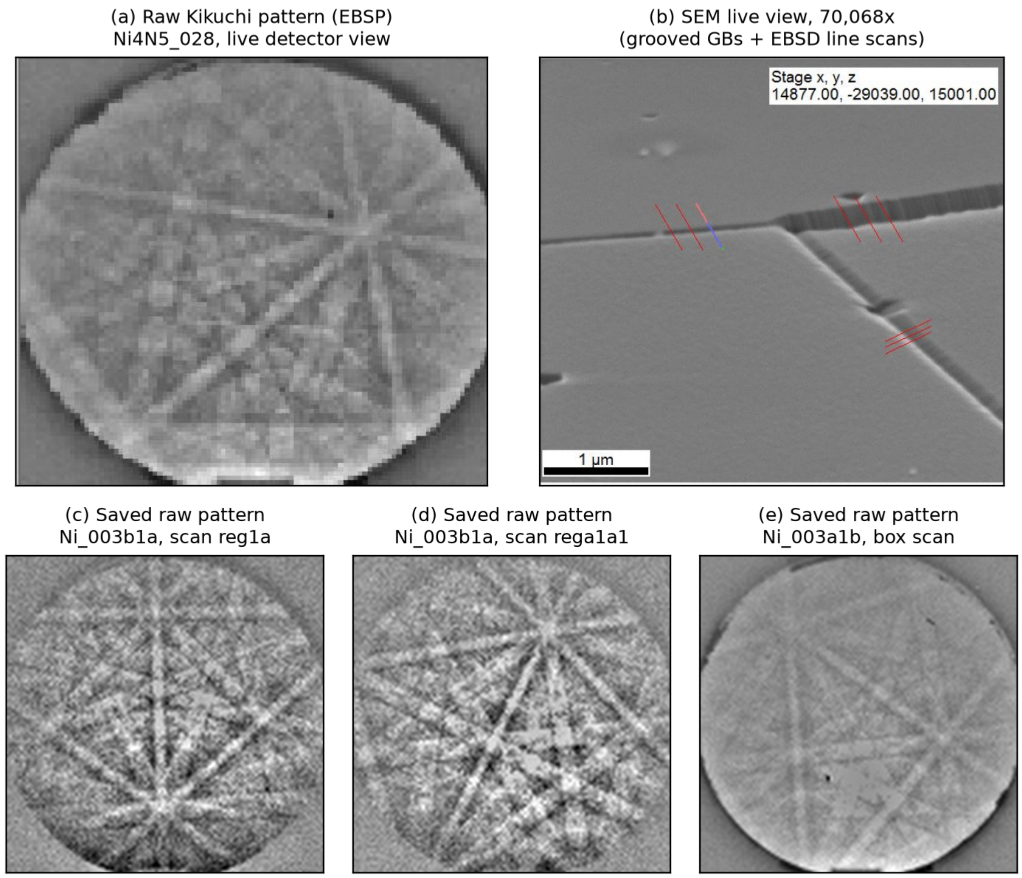

Figure S8: Raw (as-collected) EBSD acquisitions from furnace-annealed nickel, with no metallographic preparation of any kind (no grinding, polishing, or etching): each specimen was pulled directly from the furnace chamber and placed directly into the EBSD chamber. (a) Raw Kikuchi pattern (electron backscatter pattern) of specimen Ni4N5_028 as seen live on the detector at 8×8 binning (30 ms exposure), with Kikuchi bands and zone axes crisply resolved; (b) the corresponding SEM live view at 70,068×, where thermal grooving alone delineates the grain boundaries crossed by the EBSD line scans (red marks; 1 μm scale bar). (c–e) Representative per-scan-point raw patterns (115×115 px at 8×8 binning) saved automatically during EBSD mapping of the early-campaign specimens Ni_003b1a and Ni_003a1b. More than 100,000 such patterns are preserved with the openly available data.

Alt text: Five grayscale panels: a raw Kikuchi diffraction pattern of intersecting bright bands, an electron micrograph with grooved grain boundaries and line-scan marks, and three small saved detector patterns.

SIV YSZ / high-temperature extension

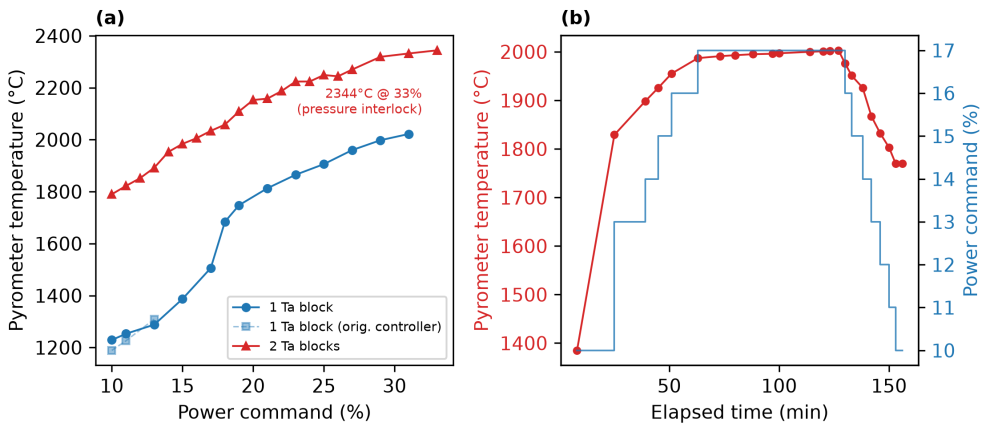

Figure S9: Measured heat curves of the tantalum-susceptor (ceramic) sample assembly. (a) Pyrometer temperature vs. generator power command for one and two stacked tantalum susceptor blocks (the faint series repeats the single-block measurement on the original power controller); the two-block ramp reached 2344 °C at a 33 % command before the chamber pressure interlock ended the test. (b) Timed two-block ramp/soak: a constant 17 % command holds approximately 2000 °C (2000 ± 3 °C over the final hour) before a stepped ramp-down.

Alt text: Two plots: temperature versus power command curves for one and two tantalum blocks reaching 2344 degrees Celsius, and a timed ramp holding a steady 2000 degrees Celsius for over an hour.

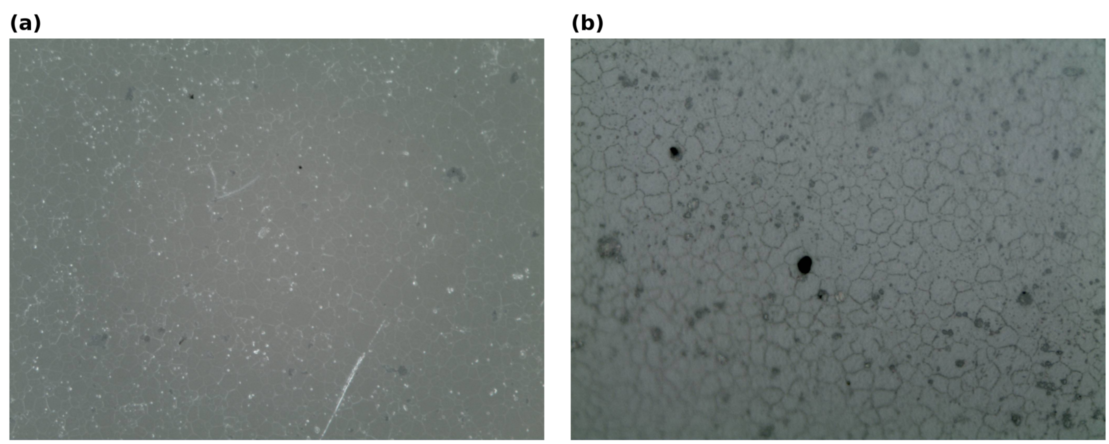

Figure S10: YSZ optical micrographs. (a) YSZ after a 1700 °C / 10 h anneal. (b) An induction-annealed YSZ specimen, showing the equiaxed grain structure. The tantalum-susceptor sample assembly used for these anneals is drawn in the main text (Fig. 8).

Alt text: Two optical micrographs of equiaxed YSZ grain structures.

SV Specimen–thermal-history linkage

Table S1 links each characterized specimen to the furnace run that produced it. Each run ID hyperlinks to the raw run log. "Nominal soak" is the target temperature for the run. "Logged soak T" is the mean pyrometer reading over the soak, taken from the recorded trace. "Atmos." indicates whether the log records active argon flow during the anneal. "Characterization" counts the recorded SEM and optical micrographs and notes where EBSD/OIM orientation data exist. The live-capture Kikuchi pattern of Fig. S8(a) was recorded on specimen Ni4N5_028. The per-scan-point patterns of Figs. S8(c)–(e) were saved during orientation mapping of the early-campaign specimens Ni_003b1a and Ni_003a1b.

Table S1: Specimen ↔ thermal-history linkage for the runs used in the manuscript's validation sections. Run IDs link to the raw run logs in the data repository.

Specimen	Material	Run (raw log)	Nominal soak	Logged soak T	Soak time	Atmos.	Characterization	Role / figure
Ni4N5_026	Ni (4N5)	IFrun039	1200 °C	1200.4 °C	6 h	no-flow	—	Calibration (Fig. S4)
Ni4N5_027	Ni (4N5)	IFrun040	1250 °C	1250.0 °C	6 h	no-flow	—	Calibration
Ni4N5_025	Ni (4N5)	IFrun038	1300 °C	1299.9 °C	1 h	no-flow	—	Calibration
Ni4N5_022	Ni (4N5)	IFrun032	1400 °C	1399.6 °C	10 min	no-flow	—	Calibration
Ni4N5_084	Ni (4N5)	IFrun079	1300 °C	1302.1 °C	12 h	flow	—	Representative soak (Fig. S5)
Ni4N5_040	Ni (4N5)	IFrun052	1200 °C	1199.8 °C	12 h	flow	—	Repeatability run 1 (Fig. S6)
Ni4N5_042,044	Ni (4N5)	IFrun054	1200 °C	1201.5 °C	12 h	flow	—	Repeatability run 2
Ni4N5_045,046	Ni (4N5)	IFrun055	1200 °C	1204.2 °C	12 h	flow	—	Repeatability run 3
Ni4N5_047,048	Ni (4N5)	IFrun056	1200 °C	1200.6 °C	12 h	flow	—	Repeatability run 4
Ni4N5_049,050	Ni (4N5)	IFrun057	1200 °C	1201.2 °C	12 h	flow	—	Repeatability run 5
Ni4N5_051,052	Ni (4N5)	IFrun058	1200 °C	1201.0 °C	12 h	flow	—	Repeatability run 6
Ni4N5_053,054	Ni (4N5)	IFrun059	1200 °C	1200.3 °C	12 h	flow	4 SEM, 3 opt. (Ni4N5_053)	Repeatability run 7; grooving (main text Fig. 5)
Ni4N5_056,057	Ni (4N5)	IFrun060	1200 °C	1200.9 °C	12 h	flow	—	Repeatability run 8
Ni200_015	Ni200	IFrun081	1325 °C	1326.5 °C	20 h	flow	3 opt.	Long soak, 20 h (Fig. S7)
Ni200_017	Ni200	IFrun080	1325 °C	1326.3 °C	40 h	flow	2 opt.	Long soak, 40 h (Fig. S7)
Ni4N5_079,080	Ni (4N5)	IFrun072	1400 °C	—	12 h	flow	—	Process-window specimen
Ni4N5_034	Ni (4N5)	IFrun049	1200 °C	1200.0 °C	12 h	no-flow	4 SEM, EBSD	EBSD IPF map (main text Fig. 4a)
Ni4N5_069	Ni (4N5)	— (log not in parsed set)	—	—	—	—	16 SEM, EBSD	EBSD IPF map (main text Fig. 4b; masked specimen)
Ni4N5_081	Ni (4N5)	IFrun082	1300 °C	1301.4 °C	20 h	flow	10 SEM, 10 opt., EBSD	Microstructure (main text Fig. 6)
Ni4N5_010	Ni (4N5)	IFrun016	1200 °C	1199.9 °C	—	no-flow	4 opt.	Characterization only
Ni4N5_012	Ni (4N5)	IFrun019	1200 °C	1199.6 °C	—	no-flow	1 opt.	Characterization only
Ni200_001	Ni200	IFrun004/IFrun005	—	—	—	no-flow	3 opt.	Characterization only
Ni4N5_001b	Ni (4N5)	IFrun006 (log not in parsed set)	—	—	—	no-flow	2 opt.	Characterization only

Alt text: Table linking each specimen to its furnace run, nominal and logged soak temperature, soak time, atmosphere, characterization file counts, and its role or figure in the manuscript.

SVI Bill of materials

Table S2 is the itemized bill of materials for the reproducible build. Designator prefixes group the parts: GEN is the induction generator package (generator, controller, chiller, heating head, and line transformer), RETRO is the reproducible retrofit (the control, vacuum, and sensing parts), and CONS is consumables. All costs are in USD and reflect the laboratory's purchase records. Several parts came from used or surplus sources as noted. Shipping and tax are excluded unless stated. The KF25 and KF40 centering-ring, clamp, and elbow counts follow the parts accounting assembled for the companion open-hardware build documentation. The GEN line items came from a single vendor quote, with the generator, heating head, controller, chiller, and line transformer totaling $16,487. For the high-temperature ceramic (YSZ) configuration, the graphite crucible assembly is exchanged for tantalum susceptor blocks resting on a ceramic stub that acts as a diffusion barrier (boron nitride for best results, sometimes MgO). Vendor records for the tantalum blocks and boron nitride stubs were not retained.

Table S2: Itemized bill of materials for the reproducible build (USD).

Desig.	Component	No.	Unit $	Total	Source	Material
GEN-1	CEIA "Power Cube" PW3-90/50 solid-state RF generator (V3000-0070)	1	$8,240	$8,240	East Coast Induction (USA)	—
GEN-2	Power Controller C-V3 Plus (V3000-0414)	1	$1,670	$1,670	East Coast Induction (USA)	—
GEN-3	Recirculating chiller (MIL043008 air-cooled closed-loop, V1650-0060; vendor-listed as a water chiller, but the loop is charged with ethylene glycol coolant)	1	$1,575	$1,575	East Coast Induction (USA)	—
GEN-4	Heating head + 3 m cable and coolant lines (PWH-13-12-30/50, V3000-0300); 1 coil made to spec at no charge with a complete unit	1	$3,337	$3,337	East Coast Induction (USA)	copper
GEN-5	Line auto-transformer, 12 kVA / 3PH / 480×380 V with taps (V049-0508; facility-dependent)	1	$1,665	$1,665	East Coast Induction (USA)	—
RETRO-1	LabVIEW-compatible DAQ with analog out + in (0–5 V AO): NI USB-6000 DAQ	1	$281.00	$281.00	National Instruments	—
RETRO-2	Dual-wavelength ratio pyrometer (LumaSense IMPAC ISR 6, 800–2500 °C), used	1	$385.00 (used)	$385.00	eBay — used listing (new list about $5,500, LumaSense quote 00161403)	—
RETRO-3	24 V linear power supply for pyrometer (International Power)	1	$51	$51	Mouser	—
RETRO-4	Voltage-to-current (0–5 V → 4–20 mA) loop conditioner set for generator power-setpoint input (NI-9265 input module + NI-9203 output module + NI cDAQ-9174 chassis)	1 set	$3,755.00	$3,755.00	National Instruments	—
RETRO-5	Edwards nEXT T-Station 85H Dry turbo pumping station (1Ph 100–120 V, 50/60 Hz, DN 40 ISO-KF)	1	$13,573.65	$13,573.65	Edwards Vacuum	—
RETRO-5A	Wide-range vacuum gauge D14701000 (36 V, 2 W); discontinued, replaced by D3G0021100	1	$1,543.29	$1,543.29	Chemtech Scientific	—
RETRO-6	Edwards TAV5 vent valve (new)	1	$652.80	$652.80	Edwards Vacuum	—
RETRO-6A	Sierra SmartTrak 100L mass flow controller (optional; helpful when continuous inert gas flow is required)	1	$550–$1,000 (quote range)	$550–$1,000	Sierra / distributor quote	—
RETRO-7	Inert-gas regulator (Fisher FS-50)	1	$38	$38	Fisher Scientific	—
RETRO-8	KF40 overpressure centering ring (STEP model in the design-file inventory, Sec. SVII)	1	$19	$19	IdealVac	aluminum
RETRO-8A	KF40 centering rings	8	$6.03	$48.24	Kurt J. Lesker	polymer / metal
RETRO-8B	KF25 centering rings	3	$3.75	$11.25	Kurt J. Lesker	polymer / metal

Continued on next page

Table S2 (continued).

Desig.	Component	No.	Unit $	Total	Source	Material
RETRO-9	KF40 vacuum clamps (one plastic quick clamp, IdealVac, $23; remainder standard clamps)	7	$6.09–$23	about $60	Kurt J. Lesker / IdealVac	aluminum / polymer
RETRO-9A	KF25 vacuum clamps	3	$5.28	$15.84	Kurt J. Lesker	aluminum
RETRO-9B	KF40 vacuum elbows (one 180°, one 90°)	2	$39.98–$89.80	$129.78	Kurt J. Lesker	stainless steel
RETRO-10	Optical window, custom quartz disc 55 mm × 1.5 mm (1 used; buy 3 for spares)	3	$18 (est.)	$54	Custom-cut metric disc — vendor not recorded in parts list; price approximate (McMaster stocks only imperial sizes)	fused quartz
RETRO-11	Ultra-high-temperature quartz disc 2 in × 1/16 in (McMaster 1357T12)	1	$19	$19	McMaster-Carr	fused quartz
RETRO-12	Inert-gas plumbing (BSPT/KF adapters, barbs, tee, overpressure relief valve)	1 set	—	about $66	McMaster-Carr / BMotionTech; the relief valve used in this build was supplied by the BYU project support center	brass / steel
CONS-1	Quartz tube, 4 ft × 35 mm ID, cut to length (a second tube as a spare is prudent but not required)	1	$67	$67	QSI Quartz	fused quartz
CONS-2	Graphite stock for machined crucible/susceptor (56L-3, 3000 °C, 1×6×6 in)	1	$99	$99	Cotronics	graphite
CONS-3	Graphite repair cement (Resbond 931-1, 3000 °C)	1	$108	$108	Cotronics	graphite
CONS-4	Alumina crucible stock (RTC-60-2, 1787 °C, 10 lb)	1	$91	$91	Cotronics	alumina
CONS-5	Zirconia crucible stock (760-1, 2204 °C, 10 lb)	1	$124	$124	Cotronics	zirconia
CONS-6	Torr-Seal high-vacuum epoxy / O-rings	1 set	$57	$57	Varian / McMaster	epoxy / elastomer

Alt text: Itemized bill of materials with designator, component description, quantity, unit and total cost in US dollars, source, and material for each generator, retrofit, and consumable part.

SVII Design-file inventory

The complete design package accompanies the raw data in the repository and the Zenodo deposit named above. The links below point directly to the files in the GitHub repository. The package comprises:

- the LabVIEW control programs — furnace control, manual control, PID tuning, and an email status alert in `code/induction-furnace-control-code/`, with the standalone manual-ramping variant in `code/manual-ramping-v5/`;
- the MATLAB heat-curve plotting script, `code/plotheatcurve.m`;
- the work-coil fabrication drawing (Fig. S2), `docs/coils-drawing.pdf`, with its editable source `docs/coils.oxps`;
- the editable system schematics, `docs/induction-furnace-schematic.pptx` and `docs/induction-furnace-schematic-v2.pptx`;
- the temperature-control wiring diagram, `docs/temp-control-modification/0-5V PLC wire design.png`;
- a STEP model of the KF40 overpressure centering ring, `docs/KF Supplies/KF40_overpressureCenteringRing.step`;
- the YSZ high-temperature documentation: the sample-assembly schematic `docs/ysz-stack-schematic.pptx`, the tantalum heat-curve workbook `docs/YSZ/Tantalum-Heat-Curve.xlsx`, the configuration survey `docs/YSZ/Induction-Furnace-Key-Takeaways.pdf`, and the chemical-compatibility notes `docs/YSZ/Observed-Adverse-Chemical-Reactions.docx`.

The control and analysis code carries an MIT license and the hardware design files carry a CERN-OHL-S license, both proposed pending final adoption. The measured dimensions of the graphite crucible/susceptor are given in Fig. S1. Exports of the LabVIEW block diagrams to PDF for readers without LabVIEW are pending.